

Advanced (Habanero)

Q1. Solve $x^2 + 5x + 6 = 0$ by factoring:

- (A) $x = -2, x = -3$
- (B) $x = -1, x = -6$
- (C) $x = 2, x = 3$
- (D) $x = 1, x = 6$
- (E) $x = -2, x = 3$

Q2. Solve $x^2 - 7x + 12 = 0$ by factoring:

- (A) $x = 3, x = 4$
- (B) $x = 5, x = 6$
- (C) $x = 3, x = 12$
- (D) $x = 4, x = 12$
- (E) $x = 3, x = 6$

Q3. Solve $x^2 + 2x - 6 = 0$ by factoring:

- (A) $x = -1\text{pm}\sqrt{7}$
- (B) $x = 1\text{pm}\sqrt{7}$
- (C) $x = -3, x = 2$
- (D) $x = -2, x = 3$
- (E) $x = -1, x = 6$

Q4. Solve $x^2 - 4x - 5 = 0$ by factoring:

- (A) $x = -1, x = 5$
- (B) $x = 1, x = 5$
- (C) $x = -5, x = 1$
- (D) $x = 5, x = -1$
- (E) $x = -1, x = -5$

Q5. Verify if $x = 2$ is a solution to $x^2 + x - 6 = 0$:

- (A) Yes
- (B) No
- (C) Maybe
- (D) Not enough info
- (E) Never

Q6. Solve $x^2 + 10x + 24 = 0$ by factoring:

- (A) $x = -4, x = -6$
- (B) $x = -2, x = -12$
- (C) $x = 4, x = 6$
- (D) $x = -4, x = 12$
- (E) $x = -2, x = 6$

Q7. Solve $x^2 - 9x + 20 = 0$ by factoring:

- (A) $x = 4, x = 5$
- (B) $x = 5, x = 6$
- (C) $x = 4, x = 20$
- (D) $x = 4, x = 10$
- (E) $x = 5, x = 4$

Q8. Solve $x^2 + x - 12 = 0$ by factoring:

- (A) $x = -4, x = 3$
- (B) $x = -3, x = 4$
- (C) $x = 4, x = 3$
- (D) $x = -4, x = 12$
- (E) $x = 3, x = 12$

Open Ended Questions (FRQ)**Q9.** Solve $x^2 - 8x + 15 = 0$ by factoring and verify your solutions.**Q10.** Solve $x^2 + 6x + 8 = 0$ by factoring and verify your solutions.**Chef's Tips**

- Make sure to check each student's work for the correct application of factoring.
- Emphasize the importance of verifying solutions.
- Encourage students to use the quadratic formula as an alternative method if factoring is not possible.